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Experimental and modeling investigation on premixed ethylbenzene flames at low pressure

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Abstract

Three premixed ethylbenzene/O_2/Ar flames with equivalence ratio (ϕ) of 0.75, 1.00, and 1.79 were studied at low pressure (4.0 kPa) to represent lean, stoichiometric, and rich ethylbenzene flames. Flame species were identified using synchrotron vacuum ultraviolet photoionization mass spectrometry, and their mole fractions were evaluated. The maximum mole fractions of hydrocarbon intermediates were observed to increase with ϕ increasing. A kinetic model including 176 species and 804 reactions was developed with detailed submechanisms of ethylbenzene and toluene. The validation of the model was made by simulating the measured mole fractions of flame species, showing good agreement in reproducing the mole fractions of most observed species. Furthermore, rate of production analysis reveals the main formation and consumption channels of some key hydrocarbon intermediates involved in ethylbenzene decomposition and PAHs formation. The main reaction channels of these species in the rich flame have salient differences with those in the lean flame, indicating the different chemistry between the pyrolysis dominated and oxidation dominated circumstance, while the chemistry in the stoichiometric flame is more similar to that in the lean flame. Furthermore, reactions involving phenyl and benzyl are concluded to be critical for PAHs formation in the rich ethylbenzene flame.
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Keywords: Premixed ethylbenzene flame; Synchrotron photoionization; Kinetic model; Ethylbenzene decomposition; PAHs formation

1. Introduction

Aromatic hydrocarbons are major components in gasolines, diesel and jet fuels. Combustion of fuels containing aromatic components is known as an important emission source of polycyclic aromatic hydrocarbons (PAHs) and soot, because the benzenoid structure of aromatic fuels facilitates the formation of these pollutants [1]. Therefore studies of aromatics combustion are critical to both pollutant control and improvement of fuel quality, and thereby attract the public concerns. Because of the moderate boiling points and soot production levels compared with PAHs, monocyclic aromatic hydrocarbons (MAHs) were broadly

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used in previous studies of aromatics combustion, particularly those applying molecular-beam sampling techniques for flame structure measurements [1].

It is known that benzene and toluene are the most studied aromatic fuels. However, increasing efforts should be devoted to the experimental and theoretical investigations on combustion of large alkylbenzenes which are often used as surrogates of diesel and jet fuels to stepwise develop their kinetic models. Ethylbenzene is the smallest aromatic hydrocarbon with a C–C bond in the side chain which is a typical structural feature of *n*-alkylbenzene surrogate components such as *n*-propylbenzene and *n*-butylbenzene. Similar to these large *n*-alkylbenzenes, ethylbenzene has a weak C–H bond on the α carbon atom and more crucially a benzylic C–C bond which is much weaker than the other types of C–C and C–H bonds in MAH molecules [2]. Therefore it has higher reactivity than benzene, toluene, and xylenes, and consequently more desirable combustion properties such as higher flame speed [3] and shorter ignition delay time [4]. Besides, recognizing the increasing sooting tendency of MAHs with longer branched chains and more complex substituted groups [5], investigation on chemical structures of ethylbenzene flames is an essential task to reveal the concentration trends of soot precursors in MAH flames and understand the formation mechanism of PAHs and soot.

In contrast to these salient advantages, only limited experimental studies have been performed on ethylbenzene combustion, including the measurements of global combustion properties such as laminar flame speeds [3], ignition delay times [4,6], and sooting tendencies [5,7,8], and the measurements of chemical structures in temperature-controlled oxidation [9,10], coflow diffusion flames [11,12] and premixed flames [13–18]. Most previous studies did not present comprehensive information of combustion intermediates, particularly the concentrations of radicals which are significant chain carriers in combustion systems. Furthermore, only one kinetic model developed from the model of benzene [19] was reported by Ergut et al. [13] to simulate premixed ethylbenzene flames at atmospheric pressure so far [13,15–17]. However, the species and reactions pertinent to the ethylbenzene submechanism were insufficiently included in their model.

In this work, three premixed ethylbenzene/O_2/Ar flames with equivalence ratios (ϕ) of 0.75, 1.00, and 1.90 were studied at low pressure (4.0 kPa) to represent the lean, stoichiometric, and rich flames. The flame species were identified using tunable synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS); and their mole fractions were evaluated. A kinetic model was developed and validated by simulating the measured mole fraction profiles of flame species in these flames. The chemical structures of the lean, stoichiometric, and rich flames provide comprehensive validation for both oxidation and pyrolysis submechanisms.

2. Experimental method

Experimental work was performed at National Synchrotron Radiation Laboratory at Hefei, China, with the detailed description of experimental instruments and experimentation published elsewhere [20–22]. The flame conditions are listed in Table 1, showing that three flames with increasing ϕ and carbon-to-oxygen (C/O) ratios were studied at 4.0 kPa. Most controllable flame conditions, including pressure, inlet percentage of Ar, and cold gas velocity of inlet mixture were kept identical in all flames for a better comparison of chemical issues. All measurements were performed along the axial direction of the burner. Methodologies of intermediate identification and mole fraction evaluation have been reported previously [21,23]. The uncertainties of evaluated mole fractions are within 10% for major species, 25% for intermediates with accurately known photoionization cross sections (PICS), and a factor of 2 for those with estimated PICS [21]. Recently, the PICS of many aromatics were measured using SVUV-PIMS [24], and were used in this work. The flame temperature was measured by using a Pt-6%Rh/Pt-30%Rh thermocouple of 0.100 mm in diameter which was coated with Y_2O_3–BeO anti-catalytic ceramic to avoid the catalytic effects [25], and was corrected for the radiation heat loss [26] and cooling effects of sampling nozzle [27]. The uncertainty of the maximum flame temperature was estimated to be $\pm$100 K.

3. Kinetic modeling

Based on the measured temperature profiles, the modeling work was performed using the CHEMKIN II and PREMIX code [28–30]. The reaction mechanism, containing 176 species and 804 reactions, mainly comprises reactions drawn from five previous mechanisms, that is, our recently reported mechanism for low pressure toluene pyrolysis [31], the USC Mech II mechanism [32] for the C_1–C_4 reactions, and the mechanisms reported by Wang and Frenklach [33], Appel et al. [34], and Richter et al. [19] for reactions involving aromatics with 1–4 rings. A detailed toluene submechanism was added to this model, which includes a series of newly studied reactions participating in toluene decomposition, such as the unimolecular decomposition reactions of toluene to produce benzyl/H and phenyl/methyl, and the H- and methyl-abstraction from toluene via attack of small radicals such as H, O, and OH

Table 1
Flame conditions.

ϕ	C/O	P/kPa	X_{Fuel}	X_{O_2}	X_{Ar}	V/cm s^{-1}
0.75	0.29	4.0	3.33	46.67	50.00	35.00
1.00	0.38	4.0	4.35	45.65	50.00	35.00
1.79	0.68	4.0	7.27	42.73	50.00	35.00

Note: X_i is the inlet percentage of species i; V is the flow velocity of inlet mixture at 300 K.

with updated rate coefficients, as listed in Table S1 in Supplemental material.

The ethylbenzene submechanism with reactions describing the decomposition of ethylbenzene was developed in this model. Table S1 presents a list of reactions in the ethylbenzene submechanism. The primary decomposition channels of ethylbenzene were proposed to be the H-abstraction to produce 1-phenylethyl ($C_6H_5CHCH_3$) and 2-phenylethyl ($C_6H_5CH_2CH_2$), the homolysis of the ethyl group to form benzyl and methyl, and the displacement or removal of the ethyl group to produce benzene or phenyl [10,11]. For the two C_8H_9 radicals, 1-phenylethyl was proposed to mainly suffer a β-scission reaction to produce styrene + H, while 2-phenylethyl have two possible β-scission reactions to produce styrene + H and phenyl + ethylene [10]. In the model reported by Ergut et al. [13], a C_8H_9 species was used to represent both 1-phenylethyl and 2-phenylethyl, and this C_8H_9 species only has one reaction, that is, the β-scission reaction to produce styrene + H. In this model, the H-abstraction and unimolecular decomposition reactions of ethylbenzene to produce both 1-phenylethyl and 2-phenylethyl, as well as the isomerization reaction between them, are included. Furthermore, the homolysis of the ethyl group, the ethyl replacement or removal, and further reactions of these primary decomposition products of ethylbenzene such as the β-scission reactions of C_8H_9 radicals to produce styrene + H and that of 2-phenylethyl to produce phenyl + ethylene were added to this model, as listed in *Table* S1. Besides, *Table* S1 also presents the reference of rate coefficients for each reaction listed.

4. Results and discussion

4.1. Flame temperature and major flame species

The flame temperature profiles and mole fraction profiles of major flame species including reactants (ethylbenzene and O_2), inert gas (Ar), and products (H_2, H_2O, CO, and CO_2) are shown in Figure 1. From Fig. 1(a), we see that the stoichiometric flame has the highest maximum temperature among the three flames, which is consistent with the fact that the stoichiometric flame has higher adiabatic flame temperature than the other two flames. Figure 1 also shows the measured and predicted mole fraction profiles of major species in the three ethylbenzene flames. The mole fractions of Ar in all flames keep dropping from the burner surface and then become almost invariable after the reaction zone, showing bending points in the mole fraction profiles. We see that this model predicted the mole fractions of Ar in fairly good agreement, and thereby successfully reproduced the mole expansion effects from reactants to products in all flames.

The predicted mole fraction profiles of reactants and major products are in good agreement with the experimental results, which provides a primary validation of this model. O_2 was depleted in the rich flame, and was only partly consumed in the lean flame, as expected. Due to the effect of thermochemical equilibrium, the stoichiometric flame also had residual O_2 in the post-flame zone. Ethylbenzene was totally consumed before the bending points or depletion positions of Ar and O_2 in all flames. The small hydrocarbon intermediates formed in ethylbenzene decomposition were sequentially destructed or oxidized and led to further mole expansion.

H_2O and CO_2 were dominant products in the post-flame zone in both lean and stoichiometric flames, while incomplete oxidation products including H_2 and CO had high concentrations in the rich flame. Also because of the effect of thermochemical equilibrium, H_2 and CO were not totally converted to H_2O and CO_2 in both lean and stoichiometric flames. It is observed that the equilibrium mole fractions of H_2 and CO in the post-flame zone increase with ϕ increasing, while the stoichiometric flame has highest equilibrium mole fractions of H_2O and CO_2 among the three flames. But if we count for the inlet hydrogen and carbon flux, respectively, the normalized equilibrium mole fractions of H_2O and CO_2 will decrease with ϕ increasing, as shown in Figure 2.

4.2. Ethylbenzene decomposition and PAHs formation

Dozens of intermediates including a lot of isomers and radicals were detected in this work. The illustrations of intermediate identification in a rich ethylbenzene flame can be found in our previous studies [23]. Table S2 in Supplemental material lists the detected intermediates, their positions and values of maximum mole fractions. As seen from this table, most intermediates with molecular weight smaller than 106 Da are identical in all flames, while the number and concentrations of aromatic intermediates with molecular weight greater than 106 Da increase as ϕ increases. It is concluded that the lean flame produces a lot of oxygenated intermediates, and the rich flame produces most PAHs among three flames. Besides, the stoichiometric flame produces moderate oxygenated intermediates and PAHs compared with

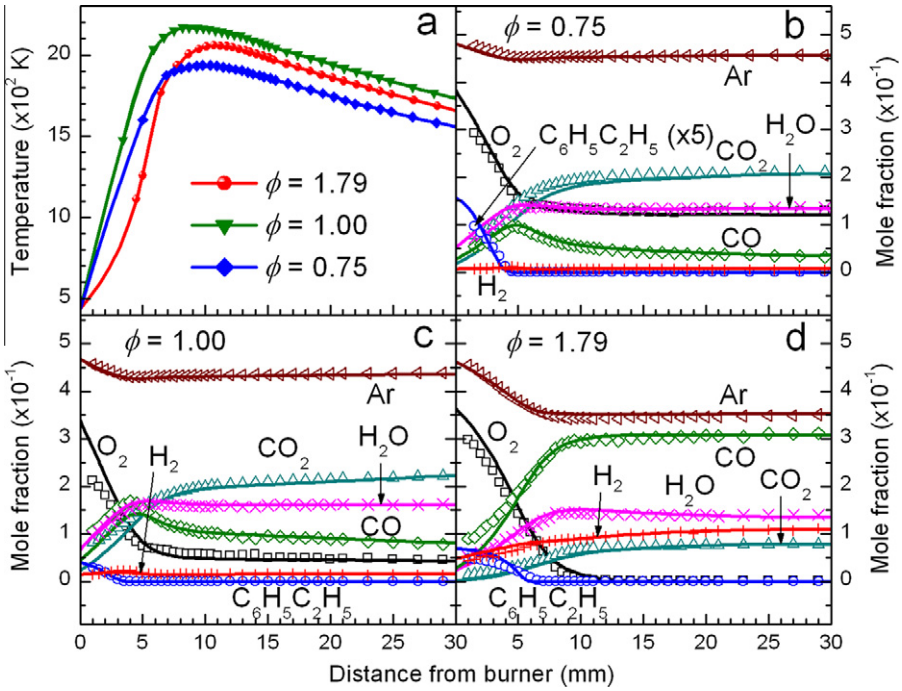

Fig. 1. (a) Temperature profiles of three ethylbenzene flames; (b)–(d) measured (symbols) and predicted (solid lines) mole fraction profiles of major species in three ethylbenzene flames, respectively.

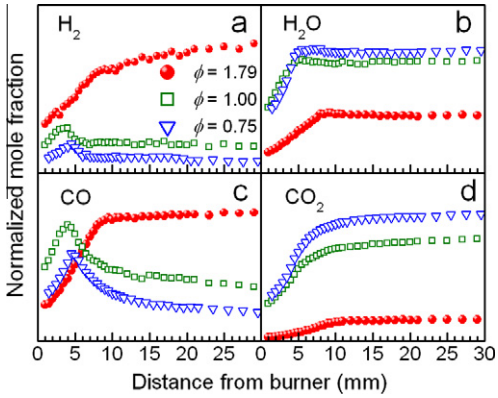

Fig. 2. Measured mole fraction profiles of (a) hydrogen, (b) water normalized by inlet hydrogen flux, (c) carbon monoxide, (d) carbon dioxide normalized by inlet carbon flux in three ethylbenzene flames.

the other two flames. The discussion in this section is mainly focused on the reduction of carbon skeleton of hydrocarbon species in all flames and the formation of typical PAHs in the rich flame.

Figure 3 shows the measured and predicted mole fraction profiles of key hydrocarbon intermediates involved in ethylbenzene decomposition process. The uncertainties of measured mole fractions are ±25% for styrene, phenylacetylene, toluene, benzene, vinylacetylene, and acetylene, and a factor of 2 for benzyl, cyclopentadienyl, and propargyl. The peak shapes of these intermediates reveal the broad reaction zones at low pressure. For a better understanding of the discussion below, Fig. 4 shows the reaction flux of ethylbenzene decomposition until cyclopentadienyl in the lean and rich flames.

Compared with benzene and toluene, ethylbenzene has more reaction types in the primary decomposition, for example, the homolysis of ethyl group to produce benzyl and methyl. The rate of production (ROP) analysis indicates that the homolysis of ethyl group is an important decomposition channel of ethylbenzene in all flames, because the C–C bond in the ethyl group has much lower bond dissociation energy than the other bonds in ethylbenzene molecule [2]. It is the dominant channel for ethylbenzene consumption in the rich flame, which is consistent with the modeling analysis of a rich ethylbenzene flame at atmospheric pressure reported by Ergut et al. [13], and consumes around one fourth of ethylbenzene in the lean and stoichiometric flames. Besides, the attack by small radicals such as H, O, and OH also has salient contributions to ethylbenzene consumption. The H attack dominates the small radical attack in the rich flame, while the attack by oxygenous radicals, particularly OH radical, has rising contributions to ethylbenzene consumption with ϕ decreasing and

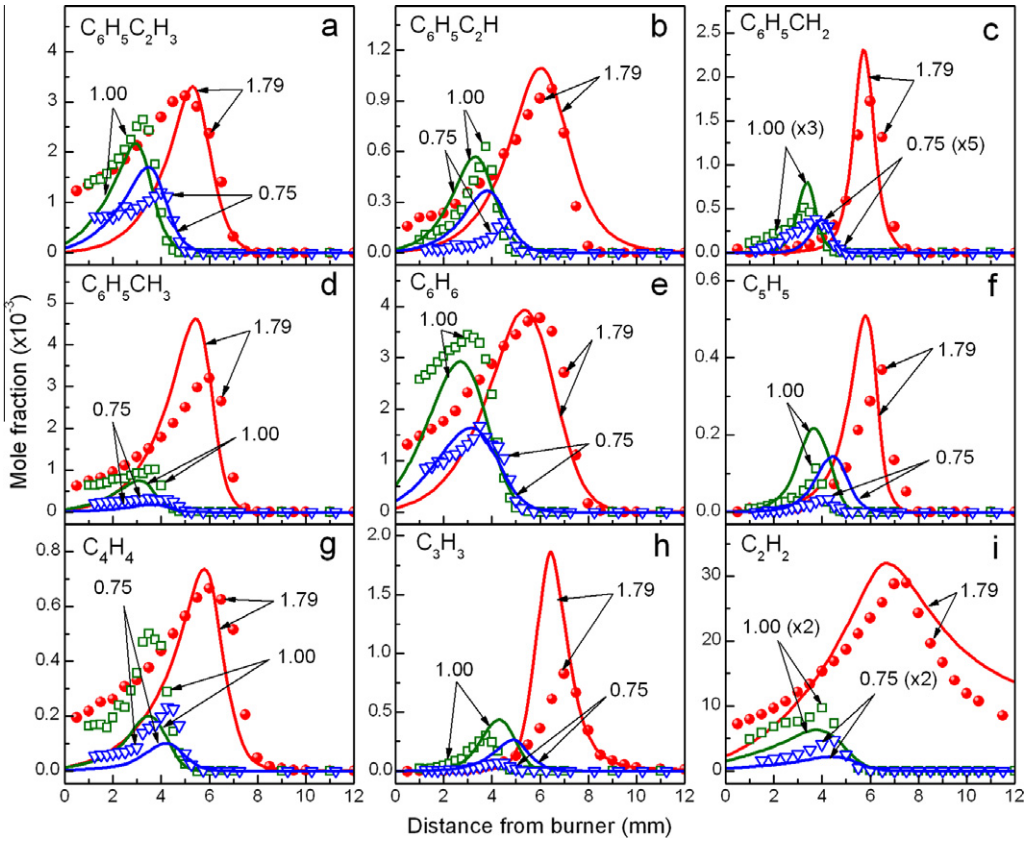

Fig. 3. Measured and predicted mole fraction profiles of (a) styrene, (b) phenylacetylene, (c) benzyl, (d) toluene, (e) benzene, (f) cyclopentadienyl, (g) vinylacetylene, (h) propargyl, and (i) acetylene in three ethylbenzene flames. Symbols and lines represent the measured and predicted mole fractions as marked with respective equivalence ratios.

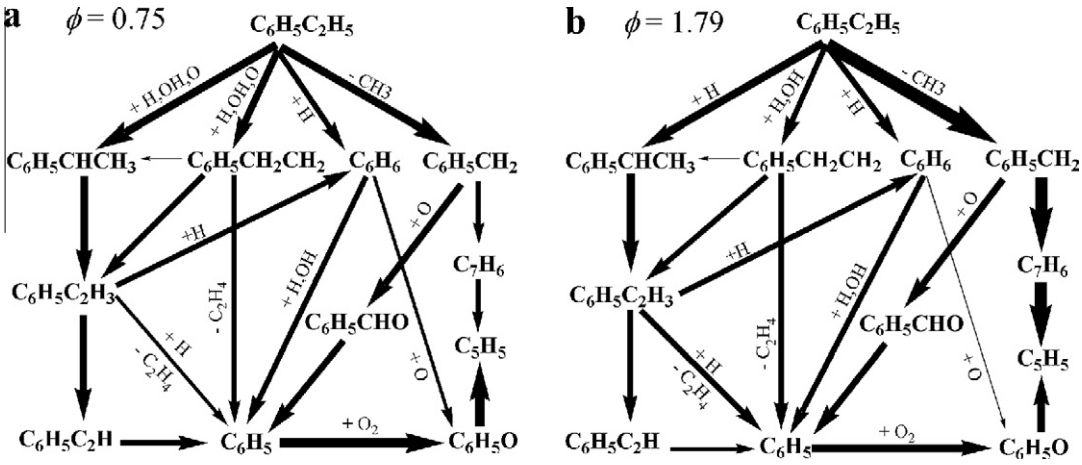

Fig. 4. Reaction flux of ethylbenzene decomposition until cyclopentadienyl over the reaction zone in (a) the lean ethylbenzene flame, and (b) the rich ethylbenzene flame.

becomes equally important to the H attack in the lean flame.

Styrene has analogous molecular structure with ethylbenzene, and was observed with higher

concentrations in the rich ethylbenzene flames than in the toluene and benzene flames with similar flame conditions in our previous studies [18,21,23,35]. In this work, its X_{max} in the rich flame is almost a magnitude high as that in a rich toluene flame at $\phi = 1.90$ [21]. The ROP analysis shows that styrene is mainly formed from the direct destruction of ethylbenzene via C_8H_9 radicals in all flames, which agrees with our previous experimental conclusion [18] and the modeling analysis reported by Ergut et al. [13]. Its destruction in the rich flame is controlled by the vinyl replacement or removal via H attack to form benzene or phenyl, and the H-abstraction reactions via H or OH attack to form C_8H_7 ($C_6H_5CCH_2$ and C_6H_5CHCH) radicals. The channels initiated by OH attack have increasing contributions as ϕ decreases, and consume nearly 40% of styrene in the stoichiometric flame and 50% in the lean flame. Similar to the formation of styrene, the formation of phenylacetylene is controlled by the destruction of styrene via C_8H_7 radicals in all flames. Its consumption mainly forms C_8H_5 radicals and phenyl in all flames. The C_8H_7 and C_8H_5 radicals are not shown in Figure 4 with consideration of length limit.

The formation of C_6 and C_7 intermediates initiates the reduction of carbon skeleton in ethylbenzene flames. Benzyl is mainly produced from the homolysis of the ethyl group as discussed above. This reaction is so fast that the X_{max} of benzyl in an ethylbenzene flame at $\phi = 1.90$ is higher than that in a toluene flame with nearly identical flame conditions [18,21]. In the rich flame, more than 40% of benzyl is destructed to form cyclopentadienyl and acetylene via fulvenallene ($C_5H_4CCH_2$) which was broadly detected in MAHs flames [18,21,23] and considered to play a significant role in benzyl decomposition [36,37]. In the lean flame, more than 50% of benzyl is converted to benzaldehyde and subsequently to phenyl. In the stoichiometric flame, the channels producing fulvenallene and benzaldehyde have almost equal contributions (around one third) to benzyl consumption, both between the contributions in the lean and rich flames. Furthermore, the reverse reaction of unimolecular decomposition of toluene is significant for benzyl depletion in all flames, and thereby governs the formation of toluene, which is in accordance with the previous modeling results reported by Ergut et al. [13].

Benzene is predominantly produced by the replacement of ethyl/vinyl group from ethylbenzene/styrene via H attack in all flames. Its destruction mainly forms phenyl in all flames. The O attack on benzene and the reaction between phenyl and O_2 lead to the formation of phenoxy which can suffer unimolecular decomposition to produce cyclopentadienyl and CO. The first channel has increasing importance as ϕ decreases and consumes almost 20% of benzene in the lean flame. As seen from Figure 3(a)–(e), the predicted mole fraction profiles of styrene, phenylacetylene, benzyl, toluene, and benzene are in good agreement with the experimental results, showing the performance of this model in reproducing the primary, secondary and tertiary decomposition steps of ethylbenzene in all flames.

Figure 3(f)–(i) exhibits the measured and predicted mole fraction profiles of cyclopentadienyl, vinylacetylene, propargyl, and acetylene. The four intermediates afford most reaction flux in the subsequent decomposition processes according to the ROP analysis. As discussed above, cyclopentadienyl plays an important role in the ethylbenzene decomposition in all flames. Its formation is governed by the H attack on fulvenallene in the rich flame, and by unimolecular decomposition of phenoxy in the lean and stoichiometric flames. Cyclopentadienyl is mainly consumed by a unimolecular decomposition reaction to produce propargyl and acetylene and also partly converted to 1,3-cyclopentadiene in the rich flame. As ϕ decreases, the oxidation of cyclopentadienyl by O attack to produce a cyclic C_5H_4O species which can be destructed to form vinylacetylene becomes important for cyclopentadienyl consumption gradually, contributing equally to the unimolecular decomposition channel in the lean flame. The further decomposition and oxidation submechanism of small molecules are almost identical in most kinetic models, and thereby will not be discussed here. As shown in Figure 3(f)–(i), the good accordance between the model prediction and experimental results validates the accuracy of this model in reactions of small molecules.

It is concluded that the main reaction channels for ethylbenzene decomposition in the rich flame have salient differences with those in the lean flame, indicating the different chemistry in the pyrolysis dominated and oxidation dominated circumstances, as shown in Figure 4. For example, the unimolecular decomposition of ethylbenzene to benzyl and methyl affords most reaction flux in the rich flame, while the reactions initiated by O and OH attack have great importance for ethylbenzene decomposition in the lean flame. Besides, the chemistry in the stoichiometric flame, though showing a transitional behavior between the lean and rich chemistry, is more similar to that in the lean flame due to the close ϕ.

The discussion of PAHs formation is focused on the results in the rich flame due to the following reasons. First, the observed PAHs have different distributions in the three flames. The heaviest molecular weight observed drops rapidly with ϕ decreasing, as seen from Table S2; the mole fractions of PAHs keep falling as ϕ decreases, because of the reduced inlet fuel concentration and more crucially the variation from pyrolysis dominated to oxidation dominated chemistry. Secondly, it is

known that the two major categories of possible mechanisms relevant to PAHs formation are the hydrogen-abstraction-carbon-addition (HACA) mechanism including sequential C_2 or C_4 species (acetylene, butadiyne, vinylacetylene, and their radicals) addition to PAH precursors [38], and the resonantly stabilized radical addition mechanism proceeded through the addition of C_3 or C_5 resonantly stabilized radicals such as propargyl and cyclopentadienyl to the PAH precursors [39,40]. These small intermediates for ring enlargement were only observed with high concentrations in the rich flame.

Figure 5 presents the measured and predicted mole fraction profiles of several typical PAHs including indene, naphthalene, acenaphthylene, and phenanthrene. The uncertainties of measured mole fractions are ±25% for indene and a factor of 2 for naphthalene, acenaphthylene, and phenanthrene. Therefore it is concluded that this model predicts the measured mole fraction profiles of these PAHs reasonably well. To guide the eye, Figure 6 shows the important formation channels of these PAHs drawn as a figure of reaction flux over the reaction zone.

The HACA channel on benzyl and the propargyl addition to phenyl dominate the formation of indene in the rich flame, producing more than 70% and 20% of indene, respectively. The consumption of indene mainly produces indenyl which is involved in the formation of larger PAHs such as acenaphthylene and phenanthrene. Many reactions contribute to the formation of naphthalene, such as the propargyl addition to benzyl, the self-combination of cyclopentadienyl, the reaction between vinylacetylene and phenyl. Among them, the propargyl addition to benzyl is the most important formation channel for naphthalene.

Acenaphthylene is the smallest tricyclic PAH. Its formation channels include the addition of propargyl or triplet propargylene to indenyl, the

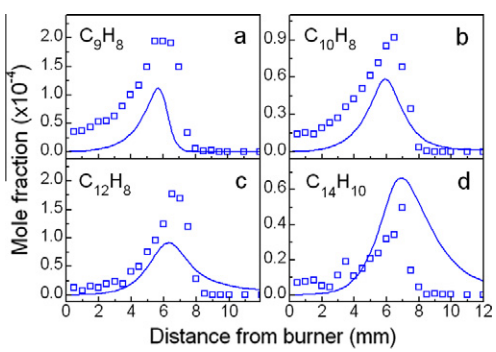

Fig. 5. Measured and predicted mole fraction profiles of (a) indene, (b) naphthalene, (c) acenaphthylene, and (d) phenanthrene in three ethylbenzene flames. Symbols and lines represent the measured and predicted mole fractions, respectively.

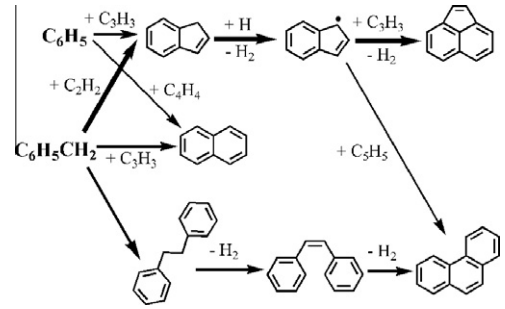

Fig. 6. Reaction flux of PAHs formation over the reaction zone in the rich ethylbenzene flame.

HACA channel on naphthyl, and so on. The ROP analysis indicates that the formation of acenaphthylene is controlled by the propargyl addition to indenyl. The oxidation reactions of acenaphthylene and its radical are important formation channels of some C_{11} aromatics. For $C_{14}H_{10}$, we observed two isomeric tricyclic PAHs, that is, phenanthrene and anthracene. Because phenanthrene is observed to be the dominant $C_{14}H_{10}$ species in the rich flame, its mole fraction profile is presented in Figure 5(d). Formation channels of phenanthrene in this model include the combination of indenyl and cyclopentadienyl, the reaction of phenylacetylene with phenyl, the HACA channels on C_{12} aromatics, and a previously proposed two-step H_2-loss channel from bibenzyl [41], etc. We see from Figure 5 that the model reasonably predicts the X_{max} of phenanthrene. Because bibenzyl can be directly formed from the self-combination of benzyl which has high concentrations in rich ethylbenzene flames, the last channel contributes predominantly to the formation of phenanthrene, while the other channels only have minor contributions. As shown in Figure 6, it is concluded that the formation of these typical bicyclic and tricyclic PAHs mainly initiates from benzyl and phenyl, revealing the important roles of the two benzenoid radicals in the formation process of PAHs.

5. Conclusions

Three premixed ethylbenzene/O_2/Ar flames with $\phi = 0.75$, 1.00, and 1.79 were studied at 4.0 kPa using SVUV-PIMS. Flame species were identified and their mole fractions were evaluated. The mole fractions of key hydrocarbon intermediates involved in ethylbenzene decomposition, as well as those of PAHs, are observed to drop with ϕ decreasing. A kinetic model with 176 species and 804 reactions was developed, including detailed submechanisms of ethylbenzene and toluene which include many recently studied reactions. The validation of the model was made by

simulating the measured mole fractions of flame species, showing reasonable agreement between the measured and predicted results. Furthermore, the ROP analysis reveals the main formation and consumption channels of the key hydrocarbon intermediates involved in ethylbenzene decomposition and PAHs formation. It is concluded that the reactions initiated from phenyl and benzyl are critical for PAHs formation in the rich ethylbenzene flame.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.159.

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